

Sijing Liu

CONTACT INFORMATION	Department of Mathematical Sciences Worcester Polytechnic Institute, Worcester, MA 01609 sliu13@wpi.edu
RESEARCH INTERESTS	Numerical analysis, Finite element methods, Discontinuous Galerkin methods, Multigrid methods, PDE-constrained optimizations, Optimal control problems, Fluid-structure interaction, Scientific machine learning
EMPLOYMENT	Worcester Polytechnic Institute, Worcester, Massachusetts , USA Post-doctoral Scholar, August 2024 – • Mentor: Professor Marcus Sarkis ICERM, Brown University, Providence, Rhode Island, USA Institute Postdoc Fellow, August 2023 – June 2024 • Mentor: Professor Johnny Guzmán University of Connecticut, Storrs, Connecticut, USA Assistant Research Professor, August 2020 – August 2023 • Mentor: Professor Dmitriy Leykekhman
EDUCATION	Louisiana State University, Baton Rouge, Louisiana, USA Ph.D., Mathematics, August 2020 • Thesis: <i>Multigrid Methods for Elliptic Optimal Control Problems</i> • Advisor: Professor Susanne C. Brenner Xiamen University, Xiamen, Fujian, China M.S., Computational Mathematics, June 2014 • Thesis: <i>On Time-Stepping Methods For Compressible Navier-Stokes Equations</i> • Advisor: Professor Chuanju Xu Fujian Normal University, Fuzhou, Fujian, China B.S., Mathematics and Applied Mathematics, June 2011
AWARDS	Travel Awards • UConn AAUP travel awards, August, 2020-August, 2023 • Math-to-industry Boot Camp IV, IMA June 23-August 02, 2019 • East Coast Optimization Meeting 2019 George Mason University (NSF supported) April 4-5, 2019 • 2018 NSF-CBMS Conference on Computational Methods in Optimal Control Jackson State University (NSF supported) July 23-27, 2018 • Industrial Math/Stat Modeling Workshop for Graduate Students SAMSI, NC State (NSF supported) July 16-26, 2017 • Frontiers in PDE-constrained Optimization, IMA June 6-10, 2016 Other Awards • Institute Postdoctoral Fellowship (NSF supported), ICERM, Brown University, 2023-2024. • SIAM Certificate of Recognition, for outstanding service and contributions to SIAM LSU student chapter, 2019. • Special Student Scholarship, Xiamen University, 2011-2014.

PUBLICATIONS

∗: Corresponding Author

Submitted:

1. SeongHee Jeong, Seulip Lee and Sijing Liu. *A monotone finite element method for an elliptic distributed optimal control problem with a convection-dominated state equation*, arXiv:2510.27167v2, 2025.
2. Sijing Liu and Jinjin Zhang. *Convergence analysis of a balancing domain decomposition method for an elliptic optimal control problem with HDG discretizations*, arXiv:2508.13997, 2025.
3. Sijing Liu[∗] and Jinjin Zhang. *A balancing domain decomposition by constraints preconditioner for a hybridizable discontinuous Galerkin discretization of an elliptic optimal control problem*, arXiv:2504.02072, 2025.

Published:

1. Satyajith Bommana Boyana, Thomas Lewis, Sijing Liu and Yi Zhang, *Convergence Analysis of a Dual-Wind Discontinuous Galerkin Method for an Elliptic Optimal Control Problem with Control Constraints*, Results in Applied Mathematics, 2025.
2. Erik Burman, Rebecca Durst, Miguel Fernández, Johnny Guzmán and Sijing Liu[∗]. *A second-order correction method for loosely coupled discretizations applied to parabolic-parabolic interface problems*, IMA Journal of Numerical Analysis, 2024.
3. Sijing Liu[∗] and Valeria Simoncini. *Multigrid preconditioning for discontinuous Galerkin discretizations of an elliptic optimal control problem with a convection-dominated state equation*, Journal of Scientific Computing, 2024.
4. Satyajith Bommana Boyana, Thomas Lewis, Sijing Liu[∗] and Yi Zhang, *Convergence analysis of novel discontinuous Galerkin methods for a convection dominated problem*, Computers & Mathematics with Applications, 2024.
5. Erik Burman, Rebecca Durst, Miguel Fernández, Johnny Guzmán and Sijing Liu[∗]. *Estimates of discrete time derivatives for the parabolic-parabolic Robin-Robin coupling method*, Numerical Algorithms, 2024.
6. Sijing Liu[∗]. *Robust multigrid methods for discontinuous Galerkin discretizations of an elliptic optimal control problem*, Computational Methods in Applied Mathematics, 2024.
7. Sijing Liu[∗], Zhiyu Tan and Yi Zhang. *Discontinuous Galerkin methods for an elliptic optimal control problem with a general state equation and pointwise state constraints*, Journal of Computational and Applied Mathematics, 2024.
8. Susanne C. Brenner, Sijing Liu and Li-yeng Sung. *Multigrid methods for an elliptic optimal control problem with pointwise state constraints*, Results in Applied Mathematics, 2023.
9. Susanne C. Brenner, Sijing Liu and Li-yeng Sung. *A P_1 finite element method for a distributed elliptic optimal control problem with a general state equation and pointwise state constraints.*, Computational Methods in Applied Mathematics, 2021.
10. Susanne C. Brenner, Sijing Liu and Li-yeng Sung. *Multigrid methods for saddle point problems: optimality systems*, Journal of Computational and Applied Mathematics, 2020.
11. Samiul Haque, Laszio P. Kindrat, Li Zhang, Vikenty Mikheev, Daewa Kim, Sijing Liu, Jooyeon Chung, Mykhailo Kuian, Jordan E. Massad, and Ralph C. Smith. *Uncertainty-enabled design of electromagnetic reflectors with integrated shape control*. In Behavior and Mechanics of Multifunctional Materials and Composites XII, vol. 10596, p. 105961D. International Society for Optics and Photonics, 2018.

In preparation:

- (with Erik Burman, Miguel Fernandez and Johnny Guzman) *Defect correction for a wave-parabolic interface problem*.
- *Discontinuous Galerkin methods for an elliptic optimal control problem with a convection-dominated state equation and pointwise state constraints*.

- (with Yi Zhang) *Dual-wind discontinuous Galerkin methods for an elliptic optimal control problem with a convection-dominated state equation.*
- (with Maurice Fabien and Marcus Sarkis) *A non-overlapping spectral additive Schwarz method for interior penalty discontinuous Galerkin discretizations of anisotropic elliptic problems.*
- (with Marcus Sarkis, Yi Zhang and Zhongqiang Zhang) *Two-scale neural networks for optimal control of convection-dominated equations.*

SPECIAL SESSION ORGANIZATION

- Special Session: Recent Advances in Numerical Methods for Partial Differential Equations and Machine Learning (with Maurice S. Fabien).
AMS 2026 Spring Eastern Sectional Meeting, Boston College, March 28-29, 2026
- Special Session: Recent Advances in Numerical Methods for Partial Differential Equations (with Yi Zhang)
AMS 2025 Spring Eastern Sectional Meeting, Hartford, April 4-5, 2025

TALKS

- Dual-wind discontinuous Galerkin methods for an elliptic optimal control problem with a convection-dominated state equation
Finite Element Circus Fall 2025, George Mason University, October 18, 2025
- Multigrid preconditioning for discontinuous Galerkin discretizations of an elliptic optimal control problem with a convection-dominated state equation
SIAM Texas-Louisiana sectional meeting 2025, Austin, September 27, 2025
- Convergence analysis of novel discontinuous Galerkin methods for a convection dominated problem
JMM 2025, Seattle, January 9, 2025
- Multigrid preconditioning for discontinuous Galerkin discretizations of an elliptic optimal control problem with a convection-dominated state equation
JMM 2025, Seattle, January 8, 2025
- A second-order correction method for loosely coupled discretizations applied to parabolic-parabolic interface problems
CSCDR Seminar, UMass Dartmouth, November 6, 2024
- Multigrid preconditioning for discontinuous Galerkin discretizations of an elliptic optimal control problem with a convection-dominated state equation
Finite Element Circus Fall 2024, UMBC, October 18-19, 2024
- A second-order correction method for loosely coupled discretizations applied to parabolic-parabolic interface problems
WPI Department of Mathematical Sciences Colloquium, October 11, 2024
- A second-order correction method for loosely coupled discretizations applied to parabolic-parabolic interface problems
9th Annual Meeting of the SIAM Central States Section
University of Missouri-Kansas City, October 5-6, 2024
- Multigrid preconditioning for discontinuous Galerkin discretizations of an elliptic optimal control problem with a convection-dominated state equation
WPI Numerical Methods Seminar, September 5, 2024
- A second-order correction method for loosely coupled discretizations applied to parabolic-parabolic interface problems
Finite Element Circus, Brown University, April 20, 2024
- Multigrid Methods,
Post Doc/Graduate Student Seminar
ICERM, Brown University, April 9, 2024
- PDE-constrained optimization,
Post Doc/Graduate Student Seminar
ICERM, Brown University, March 19, 2024
- Discontinuous Galerkin methods for an elliptic optimal control problem with a

- general state equation and pointwise state constraints,
Finite Element Circus, Bridgewater State University, March 17-18, 2023
- An Introduction to Multigrid Methods,
SIGMA Seminar, University of Connecticut, February 24, 2023
- Robust Multigrid Methods for an Elliptic Optimal Control Problem based on
Discontinuous Galerkin Methods,
JMM 2023, Boston, January 5, 2023
- Multigrid Methods for an Elliptic Optimal Control Problem based on Discontinuous
Galerkin Methods,
Finite Element Circus, Carnegie Mellon University, October 21, 2022
- FEM and Multigrid Methods for Elliptic Optimal Control Problems,
Bridgewater State University Mathematics Seminar, April 28, 2022
- Multigrid Methods for Elliptic Optimal Control Problems,
UNC Greensboro Applied Mathematics Seminar, November 15, 2021
- A P_1 Finite Element Method for a Distributed Elliptic Optimal Control Problem
with a General State Equation and Pointwise State Constraints,
UNC Greensboro PDE Conference July 24, 2021
- Multigrid Methods for Elliptic Optimal Control Problems,
UConn Control and Optimization Seminar April 12, 2021
- P_1 Finite Element Methods for a General Optimal Control Problem with Pointwise
State Constraints,
Finite Element Circus Spring 2021(Virtual) April 10, 2021
- Multigrid Methods for Elliptic Optimal Control Problems,
20th Copper Mountain Conference On Multigrid Methods (Virtual)
March 29, 2021
- Multigrid Methods for an Optimal Control Problem with Pointwise State Constraints,
Finite Element Circus Fall 2019, Virginia Tech November 2, 2019
- Multigrid Methods for an Optimal Control Problem with Pointwise State Constraints,
Scientific Computing Around Louisiana 2019 (SCALA 2019)
Tulane University February 16, 2019
- Multigrid Methods for an Optimal Control Problem,
Finite Element Circus Fall 2017, UMBC October 21, 2017
- Multigrid Methods for an Optimal Control Problem,
Scientific Computing Around Louisiana 2017 (SCALA 2017)
Tulane University March 17, 2017

WORKSHOPS

- From Modeling to Learning with HPC,
ICERM, Brown University September 13-14, 2025
- Computational Learning for Model Reduction,
ICERM, Brown University January 6-10, 2025
- Nonlocality: Challenges in Modeling and Simulation,
ICERM, Brown University April 15-19, 2024
- PDEs and Geometry: Numerical Aspects,
ICERM, Brown University March 11-15, 2024
- Numerical Analysis of Multiphysics Problems,
ICERM, Brown University February 12-26, 2024
- CMAI Meets Industry Symposium (Virtual)
George Mason University June 25, 2021
- CMAI Summer School: An Introduction to Risk-Averse PDE-Constrained
Optimization: Theory, Numerical Solution, and Open Problems (Virtual)
George Mason University June 18, 2021
- ICERM Virtual Workshop: Mathematical and Computational Approaches for
Solving the Source-Free Einstein Field Equations October 5-9, 2020
- Math-to-industry Boot Camp IV, IMA June 23-August 02, 2019

Project: Rail Car Supply Forecasting.

- 1st LBRN-LONI Scientific Computing Bootcamp, LSU May 28-29, 2018
- Industrial Math/Stat Modeling Workshop for Graduate Students, SAMSU, NC State July 16-26, 2017

Project: Uncertainty-enabled design of electromagnetic reflectors with integrated shape control.

- Frontiers in PDE-constrained Optimization, IMA June 6-10, 2016
- Nvidia GPU Workshop, LSU June 1-2, 2016
- 5th Annual LONI HPC Parallel Programming Workshop, LSU May 30-31, 2016

CONFERENCES

- Finite Element Circus 2025 Fall, George Mason University October 17-18, 2025
- SIAM TXLA Sectional Meeting 2025, Austin, September 26-28, 2025
- AMS 2025 Spring Eastern Sectional Meeting, Hartford April 4-5, 2025
- JMM 2025, Seattle January 8, 2025
- Finite Element Circus 2024 Fall, UMBC October 18-19, 2024
- 9th Annual Meeting of the SIAM Central States Section University of Missouri-Kansas City October 5-6, 2024
- Mathematical Models and Numerical Methods for Multiphysics Systems University of Pittsburgh May 1-3, 2024
- Finite Element Circus 2024 Spring, Brown University April 19-20, 2024
- Finite Element Circus 2023 Spring Bridgewater State University March 17-18, 2023
- Joint Mathematics Meeting 2023, Boston January, 2023
- Finite Element Circus 2022 Fall Carnegie Mellon University October, 2022
- SIAM Annual Meeting, Pittsburgh July 13-15, 2022
- East Coast Optimization Meeting 2022 (Virtual) George Mason University March 31-1, 2022
- Finite Element Circus 2021 Fall The Pennsylvania State University November 5-6, 2021
- UNC Greensboro PDE Conference of 2021 (Virtual) July 24-25, 2021
- Finite Element Circus 2021 Spring (Virtual) April 9-10, 2021
- East Coast Optimization Meeting 2021 (Virtual) George Mason University April 1-2, 2021
- 20th Copper Mountain Conference On Multigrid Methods (Virtual) March 29-April 2, 2021
- Finite Element Circus 2020 Fall (50th Anniversary) (Virtual) November 6-7, 2020
- IMSI Opening Conference: Vistas in the Applied Mathematical Sciences (Virtual) October 7-9, 2020
- SIAM Annual Meeting 2020 (Virtual) July 6-17, 2020
- Scientific Computing Around Louisiana 2020 (SCALA 2020) Louisiana State University February 7-8, 2020
- Finite Element Circus 2019 Fall, Virginia Tech November 1-2, 2019
- East Coast Optimization Meeting 2019 George Mason University April 4-5, 2019
- Scientific Computing Around Louisiana 2019 (SCALA 2019) Tulane University February 15-16, 2019
- Finite Element Circus 2018 Fall, University of Delaware November 9-10, 2018
- Celebrating 75 Years of Mathematics of Computation ICERM, Brown University November 1-3, 2018
- 2018 NSF-CBMS Conference on Computational Methods in Optimal Control

	- Jackson State University July 23-27, 2018 • Finite Element Rodeo 2018, Louisiana State University February 23-24, 2018 • Scientific Computing Around Louisiana 2018 (SCALA 2018) Louisiana State University February 2-3, 2018 • Finite Element Circus 2017 Fall, UMBC October 20-21, 2017 • International Conference on Current Trends and Challenges in Numerical Solution of Partial Differential Equations, Purdue University July 7-8, 2017 • Scientific Computing Around Louisiana 2017 (SCALA 2017) Tulane University March 17-18, 2017 • Scientific Computing Around Louisiana 2016 (SCALA 2016) Louisiana State University February 12-13, 2016 • International Conference on Mathematical Modeling, Analysis and Computation Xiamen, China July 21-25, 2012
TEACHING EXPERIENCES	Instructor August 2024 to present Department of Mathematical Sciences, Worcester Polytechnic Institute – MA1024 Calculus IV – MA2051 Ordinary Differential Equations – MA3257 Numerical Methods for Linear and Nonlinear Systems – MA4222 Top Algorithms in Applied Mathematics Instructor August 2020 to August 2023 Department of Mathematics, University of Connecticut – MA2410Q Elementary Differential Equations – MA3510 Numerical Analysis I – MA3511 Numerical Analysis II Teaching Assistant\Instructor August 2014 to May 2020 Department of Mathematics, Louisiana State University – MATH1021 College Algebra – MATH1550 Differential and Integral Calculus Teaching Assistant June 2011 to May 2014 School of Mathematical Sciences, Xiamen University
PROFESSIONAL MEMBERSHIPS	• American Mathematical Society • Society for Industrial and Applied Mathematics
REFeree	• Applied Mathematics and Optimization • Journal of Computational and Applied Mathematics • Results in Applied Mathematics • Science China: Mathematics • Journal of Numerical Mathematics • Computers & Mathematics with Applications • Applied Numerical Mathematics • Numerical Methods for Partial Differential Equations

SERVICE	Secretary, LSU SIAM Student Chapter	2016
	Vice President, LSU SIAM Student Chapter	2017
	President, LSU SIAM Student Chapter	2018, 2019
	Session Chair, UNC Greensboro PDE Conference	2021
	Judge, AWM Poster Session at JMM 2023	2023